

From Cross-Sections to Iterated Integrals

In Calc I you computed volumes by integrating cross-sectional areas:

$$V = \int_a^b A(x) dx$$

The key was knowing the shape of each cross-section. Let's revisit this idea with a concrete example.

Example 1

A solid S has its base in the xy -plane, bounded by $f(x) = x(x - 2)$ and $g(x) = x$. Cross-sections perpendicular to the x -axis are **squares**. Find the volume.

Find where the curves intersect, then compute the side length of each square cross-section as $g(x) - f(x)$.

The Cross-Section Idea for Surfaces

Now instead of a solid with known cross-sections, suppose we want the volume under a **surface** $z = f(x, y)$ over a rectangle $R = [a, b] \times [c, d]$.

Fix a value of x . Slicing the surface at that x gives a curve $z = f(x, y)$ in the yz -plane. The area under that curve is:

$$A(x) = \int_c^d f(x, y) dy$$

Then the total volume is:

$$V = \int_a^b A(x) dx = \int_a^b \left[\int_c^d f(x, y) dy \right] dx$$

This is an **iterated integral**: we integrate in y first (treating x as a constant), then integrate the result in x .

We write this more compactly as:

$$\int_a^b \int_c^d f(x, y) dy dx$$

The **inner integral** is computed first, then the **outer integral**.

An iterated integral is just two single-variable integrals done one at a time. The inner integral produces a function of the outer variable, and then we integrate that.

Example: Volume Under $f(x, y) = 1 + x^2 + y^2$

Example 2

Find the volume under $f(x, y) = 1 + x^2 + y^2$ above the unit square $R = [0, 1] \times [0, 1]$.

Fix x and integrate in y first to get $A(x)$. Then integrate $A(x)$ from 0 to 1.

Fubini's Theorem

Does the order of integration matter? Can we integrate in x first, then y ?

Theorem: Fubini's Theorem

If f is continuous on the rectangle $R = [a, b] \times [c, d]$, then

$$\iint_R f(x, y) dA = \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy$$

Both orders of integration give the **same answer**.

Let's verify with our previous example. Integrate in x first:

$$\int_0^1 \int_0^1 (1 + x^2 + y^2) dx dy = \int_0^1 \left[x + \frac{x^3}{3} + y^2 x \right]_0^1 dy = \int_0^1 \left(\frac{4}{3} + y^2 \right) dy = \frac{4}{3} + \frac{1}{3} = \frac{5}{3} \checkmark$$

Fubini's Theorem says we can choose whichever order is more convenient. Sometimes one order is much easier than the other!

Quick Check: Set Up an Iterated Integral

Suppose $R = [0, 2] \times [1, 3]$.

Write $\iint_R xy dA$ as an iterated integral in both orders. Then evaluate one of them.

Identify a, b, c, d from R . Then write the integral both ways: $dy dx$ and $dx dy$.

Example: A Trig Integral

Example 3

Evaluate $\iint_R \cos(x + 2y) dA$, where $R = [0, \pi] \times [0, \pi/2]$.

Integrate in y first using substitution $u = x + 2y$.

Beyond Rectangles: General Regions

So far we've integrated over **rectangles** $[a, b] \times [c, d]$, where the limits of integration are all constants. But most real-world regions aren't rectangles!

What changes? When the region is not a rectangle, the limits of the **inner integral** depend on the **outer variable**.

- On a rectangle: $\int_a^b \int_c^d f(x, y) dy dx$ — all limits are constants
- On a general region: $\int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$ — inner limits are **functions**

To set up these integrals correctly, we need to describe the region in a systematic way. That's where **Type 1** and **Type 2** regions come in.

The outer limits are always constants (numbers). The inner limits can be functions of the outer variable. Getting these limits right is the central skill of this section.

Type 1 and Type 2 Regions

Type 1 Region (Integrate dy, dx)

A region D is **Type 1** if it lies between two curves $y = g_1(x)$ and $y = g_2(x)$:

$$D = \{(x, y) \mid a \leq x \leq b, \quad g_1(x) \leq y \leq g_2(x)\}$$

$$\iint_D f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$$

Type 2 Region (Integrate dx, dy)

A region D is **Type 2** if it lies between two curves $x = h_1(y)$ and $x = h_2(y)$:

$$D = \{(x, y) \mid c \leq y \leq d, \quad h_1(y) \leq x \leq h_2(y)\}$$

$$\iint_D f(x, y) \, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) \, dx \, dy$$

Type 1: Outer variable is x , inner limits are functions of x . Integrate dy first.

Type 2: Outer variable is y , inner limits are functions of y . Integrate dx first.

Many regions are **both** types — you get to choose!

Quick Check: Identify the Limits

Consider the triangular region with vertices $(0, 0)$, $(1, 0)$, and $(0, 1)$.

Describe this region as both Type 1 and Type 2, and write the limits of integration for each.

Sketch the triangle. Find the equation of the hypotenuse. Then describe using “for a fixed x , y goes from $\dots$ to $\dots$ ”

Example: Integral over a Triangular Region

Example 4

Evaluate $\iint_D x \sqrt{y^2 - x^2} \, dA$, where $D = \{(x, y) \mid 0 \leq y \leq 1, 0 \leq x \leq y\}$.

First, sketch the region. D is the triangle below $y = x$ (equivalently, to the right of $x = 0$) with $0 \leq y \leq 1$.

This is a Type 2 region: for fixed y , x goes from 0 to y . Set up the integral as $dx \, dy$.

Describing a Region Both Ways

Example 5

Consider the region D between $y = x^2$ and $y = \sqrt{x}$.

(a) Describe D as both a Type 1 and a Type 2 region.

(b) Evaluate $\iint_D (2x - 3y^2) dA$.

First, find the intersections: $x^2 = \sqrt{x} \Rightarrow x^4 = x \Rightarrow x(x^3 - 1) = 0 \Rightarrow x = 0$ or $x = 1$.

Sketch the region. For Type 1: fix x , find y -bounds. For Type 2: fix y , find x -bounds. Then choose the easier order to integrate.

Changing the Order of Integration

Sometimes you **must** change the order of integration because the inner integral is impossible to evaluate directly.

Strategy:

1. **Sketch the region** described by the original limits
2. Re-describe it using the **other type**
3. Write the integral with the new limits

$\begin{ex}{6}$

Evaluate $\int_0^1 \int_x^1 e^{y^2} dy dx$.

$\end{ex}$

We can't integrate e^{y^2} in y — there's no elementary antiderivative! Sketch the region, then swap the order.

Homework

Section 15.2: 3, 5, 7, 9, 11, 15, 16, 17, 19, 23, 25, 27, 31, 33, 35, 37, 43, 55, 57, 59, 61, 63, 65, 75, 77